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H2O+: An Improved Framework for Hybrid Offline-and-Online RL with Dynamics Gaps

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***Abstract*— Solving real-world complex tasks using reinforcement learning (RL) without high-fidelity simulation environments or large amounts of offline data can be quite challenging. Online RL agents trained in imperfect simulation environments can suffer from severe sim-to-real issues. Offline RL approaches although bypass the need for simulators, often pose demanding requirements on the size and quality of the offline datasets. The recently emerged hybrid offline-and-online RL provides an attractive framework that enables joint use of limited offline data and imperfect simulator for transferable policy learning. In this paper, we develop a new algorithm, called H2O+, which offers great flexibility to bridge various choices of offline and online learning methods, while also accounting for dynamics gaps between the real and simulation environments. Through extensive simulation and real-world robotics experiments, we demonstrate superior performance and flexibility of H2O+ over advanced cross-domain online and offline RL algorithms.**

I. Introduction

The past successes of reinforcement learning (RL) are primarily restricted to single-domain tasks with the same environment dynamics during the training and testing phases [1], [2]. However, it has been observed that most RL algorithms are highly vulnerable to changes in environment dynamics [3]–[5], resulting in suboptimal policy performance and limiting the broader success of RL in real-world tasks. In robotics applications [6]–[9], for instance, we typically train control policies in simulators for the sake of training efficiency and safety considerations. However, the dynamics modeling within the simulator can be hard to strictly align with the diverse and complex real-world scenarios, leading to severe performance degradation due to dynamics mismatch [10]–[12].

To address sim-to-real transfer issues [13], recent RL methods have adopted several design paradigms. System identification methods [14]–[18] aim to calibrate and align the physical properties in simulation with those in the real world. Domain randomization techniques [10], [12], [19], [20] randomize simulation parameters to generalize policies across multiple environments. However, the selection of parameters and the range of their randomization could require a great amount of human effort and domain expertise [9], [21], as well as sufficient configurability of the simulator [22]. Thus, another avenue of works [4], [23] regard simulators as black boxes and turn to perform policy learning adaptation via modifying the reward to account for the sim-to-real dynamics gap. More recently, the rapid developments in offline RL [24]–[31] have brought renewed interest in learning policies directly from pre-collected real-world datasets to bypass the need for simulation environments. These methods adopt conservative principles to overcome the notorious distributional shift issue [26] in offline learning, thus often requiring large, high state-action space coverage, and high-quality datasets to achieve good performance [32] which can be hard to satisfy in scenarios with high data collection costs.

All previous approaches bear some limitations, suggesting that solely relying on online simulation samples with imperfect dynamics or limited, low-coverage real-world offline data may not be sufficient to achieve desirable policy transferability. To this end, dynamics-aware hybrid offline-and-online RL (H2O) [5] is the first study to combine offline and online policy learning using both limited offline real-world data and off-dynamics online simulated samples for cross-domain policy learning. It introduces a dynamics-aware value regularization scheme that boosts Q-values on offline real data and punishes Q-values on simulation samples based on explicit dynamics gap quantification. Although promising, H2O bears several drawbacks. First, it is built on the conservative Q-learning (CQL) [33] framework, which is over-conservative and lacks flexibility to be extended to stronger and less conservative offline RL paradigms. Its over-conservative design hinders sufficient exploration and state-action coverage improvement in the simulation environment. Lastly, explicit dynamics gap quantification in H2O also poses computation challenges.

In this paper, we follow the offline-and-online RL recipe in H2O, but develop a more flexible and powerful algorithm through a different lens, to enable sufficient utilization of both the offline dataset and imperfect simulator for transferable policy learning. We refer our algorithm as **H2O+**, which has two favorable design ingredients: 1) a *flexible and less conservative learning framework* that is compatible with various strong in-sample learning offline RL backbones and exploration designs; and 2) the *dynamics-aware mixed value update* that bridges offline and online value function learning, while also accounting for dynamics gaps between real and simulated samples. Through extensive simulation and real wheel-legged robot experiments, we demonstrate the superiority and flexibility of H2O+ over competing online, offline, and cross-domain RL baseline methods.

II. Related Work

A. Reinforcement Learning with Imperfect Simulators

High-fidelity simulators are crucial for online RL methods to learn deployable policies. However, as accurate simulators are hard to build, addressing the sim-to-real gaps become

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a pressing challenge. Various cross-domain online RL approaches have been proposed to tackle this challenge, such as using system identification methods [15]–[18], [34] to align simulated dynamics with real dynamics, or adding domain randomizations [9], [10], [12], [19], [20] that trains RL policies in a randomized simulated dynamics setting. The former typically requires a considerable amount of offline or costly real-world interaction data [14], while the latter necessitates manually-specified randomized parameters [21]. Recently, another line of research leverages additional real-world data to mitigate the dynamics shift in simulation environments [4], [5], [23]. Specifically, DARC [4] and DARA [23] add dynamics-gap-related penalization terms on rewards in online and offline RL settings, respectively. H2O [5] proposes a new setting that enables simultaneous offline-and-online policy learning on both real offline data and simulated samples, which shows promising results and advantages over prior methods.

B. Policy Learning by Combining Offline and Online RL

The recently emerged offline RL methods [25]–[31] has provided an attractive solution to learn policies directly from offline data without online interactions. However, the performances of existing offline RL methods are heavily limited by the quality and state-action space coverage of offline datasets [26], [32]. To mitigate this issue, offline-to-online RL methods [35]–[37] are developed to separate RL policy learning into a two-stage training process: first pretrain a policy using offline RL and then finetune with online RL. It can improve sample efficiency with favorable initialization for the online learning stage. More recently, some RL studies [38]–[41] directly merge offline RL ingredients into online RL algorithms as a single-stage learning process, which have been shown to greatly improve sample efficiency and policy performance. However, all these methods are only applicable to a single domain, with no dynamics gaps between the online and offline data. H2O [5] also adopts simultaneous offline and online learning, but is specifically designed to tackle off-dynamics online samples from an imperfect simulator. Our proposed H2O+ follows the same hybrid offline-and-online RL setting, but uses a different methodological framework to address several key drawbacks of H2O.

III. Preliminaries

A. Reinforcement Learning

We consider the RL problem formulated as a Markov Decision Process (MDP) [42], defined by a tuple $\mathcal{M} := (\mathcal{S}, \mathcal{A}, r, P_{\mathcal{M}}, \gamma)$. $\mathcal{S}$ and $\mathcal{A}$ denote the state and action space, r represents the reward function, $P_{\mathcal{M}}$ stands for the transition dynamics under $\mathcal{M}$. The goal of RL is to find the optimal policy π^* that maximizes cumulative discounted reward starting from an initial state distribution ρ, *i.e.*, $\pi^* = \arg\max_\pi \mathbb{E}_{\mathbf{s}_0 \in \rho, \mathbf{a}_t \sim \pi(\cdot|\mathbf{s}_t), \mathbf{s}_{t+1} \sim P_{\mathcal{M}}(\cdot|\mathbf{s},\mathbf{a})} \left[\sum_{t=0}^{\infty} \gamma^t r\left(\mathbf{s}_t, \mathbf{a}_t\right)\right]$. RL methods based on approximated dynamic programming typically learn an action-value function $Q(s,a)$, and optionally, a state value function $V(s)$ to practically estimate the cumulative discounted reward for policy optimization.

In many cases, RL training in the real environment is infeasible, so most online RL methods train the agents in simulation environments. However, building a high-fidelity simulator can be costly or even impossible in many real-world tasks. Learning with an imperfect simulator will lead to a MDP $\widehat{\mathcal{M}} := \left(\mathcal{S}, \mathcal{A}, r, P_{\widehat{\mathcal{M}}}, \gamma\right)$ with biased dynamics $P_{\widehat{\mathcal{M}}}$, which can cause serious sim-to-real transfer issues. When a large offline real-world dataset $\mathcal{D}$ generated by some behavior policy μ is given, one can also resort to offline RL [24], [25] to bypass the sim-to-real issue and directly learn a policy from the offline data. However, the performances of existing offline RL methods are heavily dependent on the size and quality of datasets, which restricts their practical application [32].

B. Hybrid Offline-and-Online RL with Imperfect Simulator

As both online and offline RL bear some practical challenges in solving real-world problems, there is a growing interest in merging online and offline RL for sample-efficient and high-performance policy learning [5], [38]–[40]. Many of these studies [38]–[40] assume identical online and offline system dynamics, thus are not applicable if an imperfect simulator is used as the online environment. Among them, H2O [5] is the first study that enables simultaneous offline and online policy learning with an imperfect simulator. H2O is built upon the conservative Q-learning (CQL) [33] framework, with its learning objective designed as follows:

$$\min_Q \underbrace{\alpha_c \cdot \left(\log \sum_{\mathbf{s},\mathbf{a}} \omega(\mathbf{s},\mathbf{a}) \exp\left(Q(\mathbf{s},\mathbf{a})\right) - \mathbb{E}_{\mathbf{s},\mathbf{a}\sim\mathcal{D}}\left[Q(\mathbf{s},\mathbf{a})\right]\right)}_{\textit{(i) Conservative value regularization}} + \tag{1}$$

$$\underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(Q - \hat{\mathcal{B}}^\pi \hat{Q}\right)(\mathbf{s},\mathbf{a})\right]^2 + \mathbb{E}_B \frac{P_{\mathcal{M}}(\mathbf{s}'|\mathbf{s},\mathbf{a})}{P_{\widehat{\mathcal{M}}}(\mathbf{s}'|\mathbf{s},\mathbf{a})}\left[\left(Q - \hat{\mathcal{B}}^\pi \hat{Q}\right)(\mathbf{s},\mathbf{a})\right]^2}_{\textit{(ii) Bellman error on offline and online data}}$$

H2O's learning objective is comprised of two parts: the first part pushes down dynamics-gap weighted Q-values and pulls up Q-values on trustworthy real offline data; the second part enables simultaneous offline and online learning on both offline dataset $\mathcal{D}$ and simulated replay buffer B while also correcting the problematic next state $\mathbf{s}'$ from the simulator dynamics $P_{\widehat{\mathcal{M}}}$ using the dynamics ratio as an importance weight. In H2O, the dynamics gap measure $\omega(\mathbf{s},\mathbf{a})$ is explicitly calculated as the normalized KL-divergence $D_{KL}(P_{\widehat{\mathcal{M}}}(\mathbf{s}'|\mathbf{s},\mathbf{a}) \| P_{\mathcal{M}}(\mathbf{s}'|\mathbf{s},\mathbf{a}))$ over all $(\mathbf{s},\mathbf{a})$ pairs in the state-action space, which can only be approximated.

IV. Method

Although H2O provides a successful attempt to tackle sim-to-real dynamics gaps by combining offline and online RL, it suffers from four notable drawbacks. First, the CQL backbone of H2O is over-conservative [32], [43] and may cause conflict when incorporating online learning. For example, as shown in [43], when performing online fine-tuning on a conservative value function initialization, policy learning has to first "unlearn" the underestimated values before making further progress. Second, the CQL framework lacks flexibility, which is not possible to be extended nor compatible with many recent strong offline RL frameworks [28], [30],

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[31], [44]. H2O also has no exploration design, which hinders effective state-action coverage improvement through simulation interactions. Lastly, H2O has to approximate an explicit dynamics gap measure, which is costly and error-prone. These drawbacks motivate us to rethink what are the desirable properties in hybrid offline-and-online RL. In this paper, we propose H2O+, which offers a highly flexible and less conservative algorithm, enabling full utilization of the online samples from the imperfect simulator. The key of H2O+ is the dynamics-aware mixed value update, which bridges various choices of offline and online learning methods, while also accounting for dynamics gaps between the real and simulation environment.

A. Separate Considerations for Offline and Online Learning

Before introducing our method, we first review two popular modeling frameworks in both offline and online RL: behavior-regularized RL (Eq.(2)) and maximum entropy RL (Eq.(3)):

$$\text{Offline: } \max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \left(r\left(\mathbf{s}_t, \mathbf{a}_t\right) - \alpha \cdot f\left(\frac{\pi\left(\mathbf{a}_t \mid \mathbf{s}_t\right)}{\mu\left(\mathbf{a}_t \mid \mathbf{s}_t\right)}\right)\right)\right] \quad (2)$$

$$\text{Online: } \max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \left(r\left(\mathbf{s}_t, \mathbf{a}_t\right) + \beta \cdot \mathcal{H}\left(\pi\left(\mathbf{a}_t \mid \mathbf{s}_t\right)\right)\right)\right] \quad (3)$$

The behavior-regularized RL is formally studied in [30], which has been shown closely related to a class of recent state-of-the-art (SOTA) in-sample learning offline RL methods [30], [44]. Depending on different choices of the f function, it can be shown that all these algorithms share the following general learning objectives for $V(\mathbf{s})$ and $Q(\mathbf{s}, \mathbf{a})$:

$$\min_{V} \mathbb{E}_{(\mathbf{s},\mathbf{a})\sim\mathcal{D}}\ \mathcal{L}_V^f(Q(\mathbf{s},\mathbf{a}) - V(\mathbf{s})) \quad (4)$$

$$\min_{Q}\ \mathbb{E}_{(\mathbf{s},\mathbf{a},\mathbf{s}')\sim\mathcal{D}}\left[r(\mathbf{s},\mathbf{a}) + \gamma V(\mathbf{s}') - Q(\mathbf{s},\mathbf{a})\right]^2 \quad (5)$$

In particular, if $f = \log(x)$, it correspond to EQL [30] and XQL [31] with $\mathcal{L}_V^f(y) = \exp(y/\alpha) - y/\alpha$. If $f = x - 1$, it corresponds to SQL [30] (equivalent to an in-sample learning version of CQL) with $\mathcal{L}_V^f(y) = \mathbb{1}(1 + y/2\alpha > 0)(1 + y/2\alpha)^2 - y/\alpha$. The well-known offline RL algorithm IQL [28] also belongs to this family of algorithms but does not have a closed-form f, with $\mathcal{L}_V^f(y) = |\tau - \mathbb{1}(y < 0)|y^2$, where $\tau \in (0, 1)$ is the expectile hyperparameter. These offline RL methods learn $V(\mathbf{s})$ and $Q(\mathbf{s}, \mathbf{a})$ completely using dataset samples, thus enjoying stable value function learning as compared to CQL-style algorithms that distort the value estimates. However, their in-sample learning nature also creates obstacles to incorporating online learning with imperfect simulators.

On the other hand, the maximum entropy RL [45] (Eq.(3)) also achieves great success in online RL studies, which maximizes the expected reward while also maximizing the entropy of the policy $\mathcal{H}(\pi)$ to promote exploration. If we consider an off-policy setting and denote B as the training replay buffer, its corresponding action-value function learning objective is given as:

$$\begin{aligned}\min_{Q}\ \mathbb{E}_{(\mathbf{s},\mathbf{a})\sim B}\big[r(\mathbf{s},\mathbf{a}) + \gamma\mathbb{E}_{\mathbf{s}'\sim P,\mathbf{a}'\sim\pi}\big[\hat{Q}(\mathbf{s}',\mathbf{a}') \\ - \beta\cdot\log(\pi(\mathbf{a}'|\mathbf{s}'))\big] - Q(\mathbf{s},\mathbf{a})\big]^2\end{aligned} \quad (6)$$

B. Dynamics-Aware Mixed Value Update

Both the behavior-regularized RL and the maximum entropy RL frameworks bear some attractive features for the hybrid offline-and-online RL setting. For example, the behavior-regularized RL provides high-quality offline learned value functions without posing too much conservatism. While the maximum entropy RL offers natural exploration capabilities to help improve the state-action space coverage of the offline dataset. Now the question is: how can we leverage the strengths of the two frameworks to build a strong hybrid RL algorithm, while also being capable of tackling the sim-to-real dynamics gaps between real and simulation environments?

In this paper, we provide a simple and elegant solution by proposing a dynamics-aware mixed value update that seamlessly mixes offline and online learning without introducing excessive conservatism. Our insight is by noting that we can use the more reliable state value function $V(\mathbf{s})$ learned solely with the real offline data in Eq.(4) as an anchor to mildly regulate $Q(\mathbf{s}, \mathbf{a})$ estimation on potentially biased simulation samples. We can achieve this by utilizing the following mixed Bellman operator [41], balancing exploitation to real-world offline data and exploration with online simulation data:

$$\begin{aligned}&\mathcal{B}_\lambda^{\text{mix}}\hat{Q}(\mathbf{s},\mathbf{a}) = \lambda\big[r(\mathbf{s},\mathbf{a}) + \gamma V(\mathbf{s}')\big] + \\ &(1-\lambda)\left[r(\mathbf{s},\mathbf{a}) + \gamma\mathbb{E}_{\mathbf{a}'\sim\pi}\left[\hat{Q}(\mathbf{s}',\mathbf{a}') - \beta\cdot\log(\pi(\mathbf{a}'|\mathbf{s}'))\right]\right] \\ &= r(\mathbf{s},\mathbf{a}) + \lambda\gamma V(\mathbf{s}') + (1-\lambda)\gamma\mathbb{E}_{\pi}\left[\hat{Q}(\mathbf{s}',\mathbf{a}') - \beta\cdot\log\pi(\mathbf{a}'|\mathbf{s}')\right]\end{aligned} \quad (7)$$

where the state value function is learned only with real-world offline data $\mathcal{D}$ as in Eq.(4); $\lambda \in [0, 1]$ is a trade-off hyperparameter to control the level of influence between offline and online learning. With the mixed Bellman operator, we can learn Q-function from both real dataset $\mathcal{D}$ and online simulation replay buffer B. Moreover, to correct potentially problematic next states $\mathbf{s}'$ from the simulator dynamics $P_{\widehat{\mathcal{M}}}$, we adopt the same dynamics ratio reweighting as in H2O [5]:

$$\begin{aligned}\min_{Q}\ &\mathbb{E}_{(\mathbf{s},\mathbf{a},\mathbf{s}')\sim\mathcal{D}}\left[\left(Q - \mathcal{B}_\lambda^{\text{mix}}\hat{Q}\right)^2(\mathbf{s},\mathbf{a})\right] + \\ &\mathbb{E}_{(\mathbf{s},\mathbf{a})\sim B}\mathbb{E}_{\mathbf{s}'\sim P_{\mathcal{M}}}\left[\left(Q - \mathcal{B}_\lambda^{\text{mix}}\hat{Q}\right)^2(\mathbf{s},\mathbf{a})\right] \\ =&\mathbb{E}_{(\mathbf{s},\mathbf{a},\mathbf{s}')\sim\mathcal{D}}\left[\left(Q - \mathcal{B}_\lambda^{\text{mix}}\hat{Q}\right)^2(\mathbf{s},\mathbf{a})\right] + \\ &\mathbb{E}_{(\mathbf{s},\mathbf{a},\mathbf{s}')\sim B}\left[\frac{P_{\mathcal{M}}(\mathbf{s}' \mid \mathbf{s},\mathbf{a})}{P_{\widehat{\mathcal{M}}}(\mathbf{s}' \mid \mathbf{s},\mathbf{a})}\left(Q - \mathcal{B}_\lambda^{\text{mix}}\hat{Q}\right)^2(\mathbf{s},\mathbf{a})\right]\end{aligned} \quad (8)$$

The dynamics ratio $P_{\widehat{\mathcal{M}}}/P_{\mathcal{M}}$ can be conveniently estimated by learning a pair of domain discriminators $p(\text{real}|\mathbf{s},\mathbf{a},\mathbf{s}')$ and $p(\text{real}|\mathbf{s},\mathbf{a})$ using the following formulation, which is also adopted in a number of previous studies [4], [5], [23]:

$$\begin{aligned}\frac{P_{\mathcal{M}}(\mathbf{s}'|\mathbf{s},\mathbf{a})}{P_{\widehat{\mathcal{M}}}(\mathbf{s}'|\mathbf{s},\mathbf{a})} &= \frac{p(\mathbf{s}'|\mathbf{s},\mathbf{a},\text{real})}{p(\mathbf{s}'|\mathbf{s},\mathbf{a},\text{sim})} = \frac{p(\text{sim}|\mathbf{s},\mathbf{a})}{p(\text{real}|\mathbf{s},\mathbf{a})}\Big/\frac{p(\text{sim}|\mathbf{s},\mathbf{a},\mathbf{s}')}{p(\text{real}|\mathbf{s},\mathbf{a},\mathbf{s}')} \\ &= \frac{1 - p(\text{real}|\mathbf{s},\mathbf{a})}{p(\text{real}|\mathbf{s},\mathbf{a})}\Big/\frac{1 - p(\text{real}|\mathbf{s},\mathbf{a},\mathbf{s}')}{p(\text{real}|\mathbf{s},\mathbf{a},\mathbf{s}')}\end{aligned} \quad (9)$$

Finally, with the learned action value function $Q(\mathbf{s}, \mathbf{a})$, we can optimize the policy π by maximizing the following objective on both real and simulated samples:

$$\pi^* = \arg\max_{\pi}\mathbb{E}_{(\mathbf{s},\mathbf{a})\in\mathcal{D}\cup B}[Q(\mathbf{s},\mathbf{a}) - \beta\cdot\log(\pi(\mathbf{s},\mathbf{a}))] \quad (10)$$

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C. Discussion and Comparison with H2O

Using the above dynamics-aware mixed value update, H2O+ effectively addresses all the aforementioned drawbacks of H2O. First, H2O+ uses in-sample learning state-value function $V(s)$ and the dynamics ratio to mildly regulate the value function learning on potentially problematic online simulated samples. There is no distortion nor extra conservative penalty on the Q-values, thus removing excessive conservatism during policy learning. Second, H2O+ is compatible with a series of recent strong in-sample learning offline RL methods [28], [30], [31], [44], and the policy entropy in Eq.(7) is also possible to be replaced with other terms to promote exploration, thus offering great flexibility. Moreover, H2O+ removes the need to estimate explicit dynamics gap measures, thus providing a simpler and more efficient algorithmic implementation. In the next section, we will show in empirical experiments, that although it removes much conservatism to regulate off-dynamics samples, H2O+ consistently outperforms H2O and other cross-domain RL baselines.

V. Experiments

In this section, we present empirical validations of our approach. We begin with our algorithmic implementation and experimental setups, followed by benchmark experiments with original and dynamics-modified MuJoCo simulation environments. Our baselines consist of the online RL method SAC [45] for zero-shot transfer, offline RL algorithms CQL [33] and IQL [28], cross-domain online RL method DARC [4], and H2O [5] that in a similar setting with H2O+. We run all experiments with 5 random seeds. Finally, we deploy H2O+ and baselines on a wheel-legged robot to complete real-world tasks. Furthermore, we provide ablations on choices of dynamics-aware mixed value update designings, different levels of intensity of dynamics gap and different offline RL backbones.

A. Experimental Setups

1) Algorithmic implementation of H2O+: In all our comparative experiments, we instantiate $\mathcal{L}_V^f(y) = |\tau - \mathbb{1}(y < 0)|y^2$ as in IQL [28], due to its simplicity. The scaling parameter β of the entropy term is automatically tuned following the treatment in SAC [45]. We follow the treatment in SAC [45] to automatically tune the scaling parameter β of the entropy term. We set the trade-off hyperparameter λ to 0.1 in all our experiments. It might be preferable to select a larger λ for tasks that are carried out in more reliable simulators.

2) Simulation experiments: We treat the original MuJoCo task environment as the "real-world" scenario, and create ten imperfect simulation environments by deliberately introducing various types of dynamics gaps (illustrated in Figure 1). These dynamics gaps are introduced by adjusting either the dynamics parameters of the robot or the environmental physical properties. For example, in the HalfCheetah and Walker2d task environment, the modifications include modifying the gravitational gravity ($\times 2$, **Gravity**), friction coefficient ($\times 0.3$, **Friction**), thigh size ($\times 0.5$ and $\times 2$, **Small/Big Thigh**), the motion range of the joint connections of thighs ($\times 0.5$ and $\times 2$, **Broken/Flexible Thigh**), stretching the torso length ($\times 4$, **Long Torso**, only for HalfCheetah), lowering the foot stiffness ($\times 0$, **Soft Feet**, only for HalfCheetah) and adding joint noise ($N(0, \mathbf{I})$, **Joint Noise**). In terms of the "real-world" offline dataset (original MuJoCo environment), we utilize the corresponding task datasets from the widely-used offline RL benchmark D4RL [46]. Specifically, the Medium (-m) and Medium Replay (-mr) datasets are considered as they are closer to real-world settings, where we are more likely to obtain medium-level or highly mixed offline datasets from real systems. The online training of algorithms is performed in the created imperfect simulation environment, and we evaluate the learned policy in the original MuJoCo environment.

3) Real-robot transfer experiments: We also perform real robot transfer experiments on a wheel-legged robot with a main body and a pair of legs with wheels attached to the end. We also construct the simulation environment based on Isaac Gym [47]. Both the real robot and its simulation are shown in Figure 2a. Our wheel-legged robot bears a substantially large weight (about 12 kg) and possesses an intricate mechanical structure. Moreover, our testing environment features a furry carpet on the ground, which introduces the possibility of sagging and unloading as the robot traverses the surface. These distinctive factors collectively contribute to a very challenging real-world transfer procedure.

According to its sensors and actuators, the state space of the robot control tasks is designed as a quadratic-tuple $(\theta, \dot{\theta}, x, v)$ where θ denotes the forward pitch angle of the body, x is the displacement of the robot, $\dot{\theta}$ and v are the angular and linear velocity respectively. The execution action is the torque τ of the motors at the two wheels. We construct two tasks for real-world validation: (1) **standing still**: the robot needs to keep balanced at the initial location and maintain stability as much as possible. The reward r is calculated as: $r = 30.0 - x^2 - v^2 - \theta^2 - \dot{\theta}^2 - \tau^2$. Ideally, when the robot standing at the original space without any swinging, the penalty item $x^2 + v^2 + \theta^2 + \dot{\theta}^2$ will be minimized. To further avoid shaking and to protect the wheel motor, we limit the wheel torque by adding a penalty τ^2 into the reward. (2) **moving forward**: the robot needs to move at a fixed forward speed $v_{tgt} = 0.2m/s$ and maintain balance as long as possible. The state space of this task $\mathbf{s} = (v, \theta, \dot{\theta})$, where the state x is no longer included as we hope that the robot can move forward in any displacement. The reward is calculated as: $r = 15.0 - (v - 0.2)^2 - \tau^2$, in which we use the penalty item $(v - 0.2)^2$ to regulate the moving speed of the robot while adding $-\tau^2$ to avoid shaking. We collect offline datasets for the standing still and moving forward tasks respectively by recording 16,588 real-world human-controlled transitions (about 90s of real-time control).

B. Comparative Results

1) Simulation experiments: The results presented in Table I highlight the superiority of H2O+ compared to all the baseline methods in terms of the mean return across all tasks in the HalfCheetah and Walker2d environments. Note that for offline

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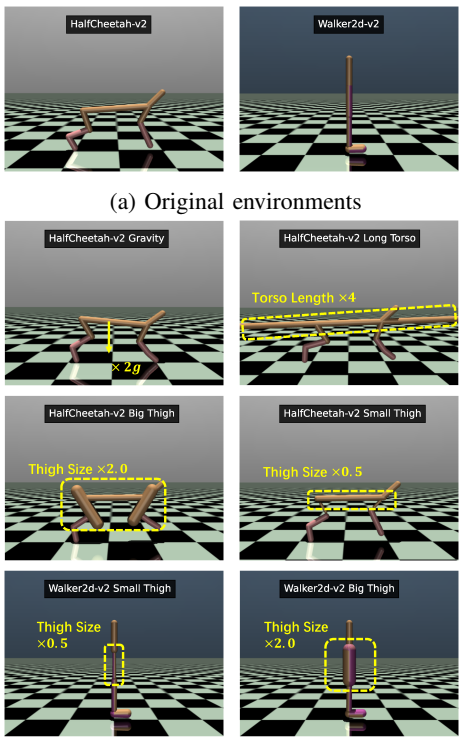

(a) Original environments

(b) Example modified environments

Fig. 1: Original environments and some illustrations of the modified dynamics

TABLE I: Average returns for MuJoCo HalfCheetah and Walker2d tasks

Data	Dynamics Gap	SAC	CQL	IQL	DARC	H2O	H2O+
HalfCheetah-mr	Gravity	4513±513	5774±214	5207±149	5105±460	**6813±289**	**6861±268**
	Friction	2684±2646	5774±214	5207±149	5503±263	5928±896	**6278±1336**
	Joint Noise	4137±805	5774±214	5207±149	5137±225	6747±427	**6985±328**
	Big Thigh	4509±877	5774±214	5207±149	5336±389	6278±305	**6675±231**
	Small Thigh	6632±1027	5774±214	5207±149	**8331±454**	6751±231	7425±148
	Broken Thigh	6517±1076	5774±214	5207±149	**8704±1726**	6717±226	7018±147
	Flexible Thigh	5623±2862	5774±214	5207±149	5554±88	6976±234	**7497±196**
	Long Torso	1047±3089	5774±214	5207±149	45±322	6225±100	**6718±245**
	Soft Feet	5684±587	5774±214	5207±149	**9058±374**	6731±319	7068±244
	Mean Return	4594	5774	5207	5863	6573	**6947**
HalfCheetah-m	Gravity	4513±513	6066±73	5605±25	5011±456	**7085±416**	**6965±659**
	Friction	2684±2646	6066±73	5605±25	6113±104	6848±445	**7186±859**
	Joint Noise	4137±805	6066±73	5605±25	5484±171	7212±236	**7503±237**
	Big Thigh	4509±877	6066±73	5605±25	6302±1832	6625±579	**7094±371**
	Small Thigh	6632±1027	6066±73	5605±25	**9127±907**	7020±337	7706±185
	Broken Thigh	6517±1076	6066±73	5605±25	**7509±707**	6800±378	7321±213
	Flexible Thigh	5623±2862	6066±73	5605±25	7266±1771	7005±757	**7805±139**
	Long Torso	1047±3089	6066±73	5605±25	724±921	**6327±602**	5484±1382
	Soft Feet	5684±587	6066±73	5605±25	6952±3330	7138±326	**7622±53**
	Mean Return	4594	6066	5605	6054	6896	**7187**
Walker2d-mr	Gravity	1698±1611	3261±802	3390±326	2969±1043	3366±740	**3518±605**
	Friction	2779±870	3261±802	3390±326	3644±213	**3916±549**	**3866±840**
	Joint Noise	173±727	3261±802	3390±326	-3±0	3045±911	**3446±862**
	Big Thigh	1151±716	3261±802	**3390±326**	57±126	1789±1781	2977±771
	Small Thigh	894±519	3261±802	3390±326	1294±905	2455±1301	**3920±417**
	Broken Thigh	**3845±607**	3261±802	3390±326	893±180	2702±1054	**3911±405**
	Flexible Thigh	2518±1627	3261±802	3390±326	2511±1048	1891±1001	**3535±493**
	Mean Return	1865	3261	3390	1624	2738	**3596**

RL baselines (CQL and IQL), we train the policies using the "real" offline datasets and evaluate them in the "real" environments, so their scores remain the same across modified environments with dynamics gaps.

It is found that online cross-domain baseline DARC performs strongly when the dynamics gap is small (i.e. when online SAC achieves better performance than offline RL baselines). However, DARC fails miserably when the dynamics gap is large (e.g. HalfCheetah long torso and Walker2d joint noise tasks), which underscores its limitations of the dynamics-gap-related reward penalization scheme. On the other hand, offline RL baselines are not impacted by the sim-to-real issue, but their performances are heavily impacted by the dataset quality. Our proposed H2O+ not only outperforms H2O in most tasks, but also consistently achieves comparable or better performance than online, offline, and cross-domain RL methods in both small and large dynamics gap settings. These results demonstrate the effectiveness of H2O+ in leveraging both offline data and imperfect simulation for improved and transferable policy learning.

2) Real-robot experiments: In the real-robot experiments, H2O+ demonstrated a strong transfer ability compared to other benchmarks. As shown in Figure 2b and 2c, the control performance of this method far exceeds that of others in both tasks. In the **standing still** task, only H2O+ and IQL policies successfully maintain the balance of the robot for over 30 seconds (s), while the robot deployed with SAC, H2O, and DARC policies cannot even keep the balance for over 3s before it hit the ground. Moreover, as illustrated in Figure 2b, H2O+ regulated the displacement of the robot within 0.2m, whereas IQL only barely maintains balance, yet with a large range swinging even reaches 1.6m. The advantage of H2O+ is more significant in the **moving forward** task. As illustrated in Figure 2c, only the H2O+ policy achieves the goal of moving forward and even follows the target velocity precisely. During the moving process, the speed of robot changes smoothly and the pitch angle remains steady. In comparison, IQL policy is capable to keep balance, but the robot moves backward and also spends more time and effort on keeping balance, resulting in a shaking period of over 7s. In addition, H2O, SAC and DARC fail to maintain balance and fall down in 4s. Additionally, as in Figure 3, we observe that H2O explores a more focused high-value area, whereas H2O+ spans a broader high-value region, thus indicating superior diversity characteristics in simulated data, which would benefit the overall performance. We also test H2O+ in a more biased simulator with an intentionally introduced mass reduction, while H2O+ still outperforms other baselines. Please see the H2O+ webpage for more details.

C. Ablation Studies

1) Ablation on the hyperparameter λ and dynamics ratio: Table II presents the results of different choices of the trade-off parameter λ and the dynamics ratio on Bellman error in dynamics-aware mixed value update, as used in Eq. 8. Specifically, we investigate the cases where $\lambda = 0$ corresponds to Q-value update on both simulation and real data using Eq. 6, and $\lambda = 1$ formulates Q-value update with

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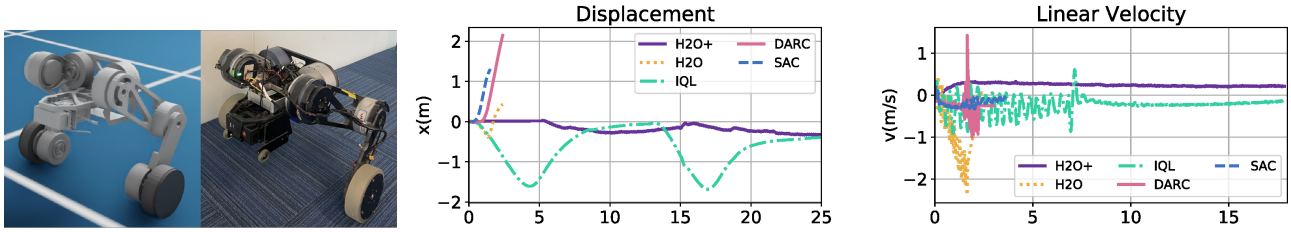

(a) Simulation/Real robot (b) Displacement in standing still (c) Linear velocity in moving forward

Fig. 2: The real-robot experiment results of the "standing still" (b) and "moving forward" (c) tasks

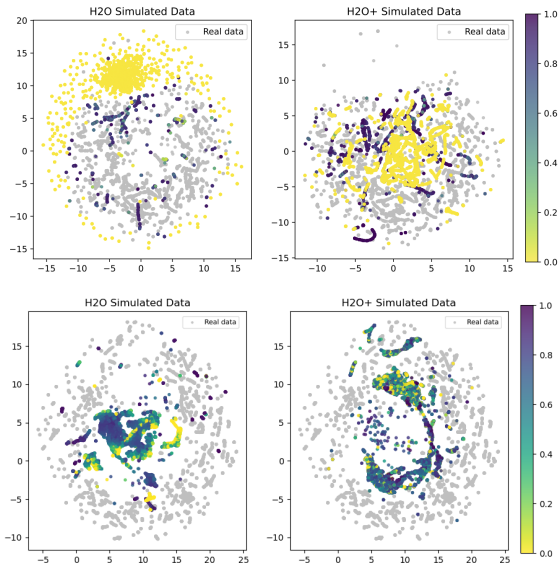

Fig. 3: Comparison of H2O / H2O+ simulation data quality in real-world tasks. (Top: standing still; Down: moving forward)

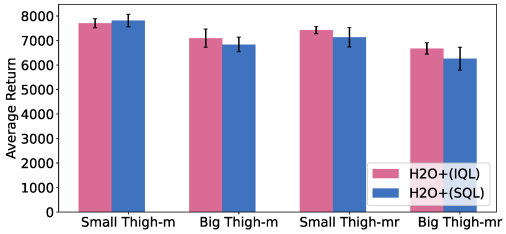

Fig. 4: Different choices of offline RL backbone for state-value function learning

Eq. 5. Among the different parameter choices, the original implementation of H2O+ with a mixed Q-value update at $\lambda = 0.1$ achieves the highest performance compared to other selections. It reveals that there is no necessity to heavily regulate the Q target by incorporating too much information from the value function learned as an anchor from offline data. Moreover, it is evident that H2O+ exhibits a remarkable level of hyper-parameter insensitivity, as indicated by its minor performance discrepancies across the λ range of 0.0 to 0.5. Essentially, our analysis also reveals that the absence of dynamics ratio reweighting in the Bellman error results in significant performance degradation.

2) Ablation on offline RL backbones: Furthermore, we plug in other offline RL backbones like SQL [30] into H2O+ paradigm for state value learning, by replacing the value loss

TABLE II: Ablations on choices in dynamics-aware mixed value update designings (λ and dynamics ratio)

Trade-off λ	0.0	0.1	0.2
Dynamics ratio	✓	✓	✓
Average return	6738±444	**6861±268**	6677±252
Trade-off λ	0.5	1.0	0.1
Dynamics ratio	✓	✓	✗
Average return	6563±752	6242±68	5579±530

TABLE III: Ablations on different levels of dynamics gap

Gravity	@1.25	@1.5	@2.0	@3.0
H2O	6846±572	6483±529	6813±289	**6171±1209**
H2O+	**7165±134**	**6948±258**	**6861±268**	**6135±811**

in Eq.(2) with $\mathcal{L}_V^f(y) = \mathbb{1}(1+y/2\alpha > 0)(1+y/2\alpha)^2 - y/\alpha$. We demonstrate H2O+ offers flexibility to bridge other offline RL algorithms in Small and Big Thigh tasks on HalfCheetah Medium and Medium Replay datasets, producing comparable performance as shown in Figure 4.

3) Investigations on different levels of dynamics gaps: We further compare H2O and H2O+ under different levels of dynamics gaps (HalfCheetah-mr with 1.25 to 3 times gravity). The results are presented in Table III. It is observed that H2O+ beats H2O in all different dynamics discrepancy levels, despite using a simpler approach to handle dynamics gaps. In addition, H2O+ consistently demonstrates a lower variance across all tasks, underscoring its heightened stability compared to H2O. Furthermore, H2O+ performs much better in tasks with low dynamics gaps and still maintains competitive performance in high dynamics scenarios. This alignment with the underlying design philosophy of leveraging the full potential of online learning with less conservatism further accentuates the superiority of H2O+.

VI. Conclusion

In this paper, we propose an improved hybrid offline-and-online RL framework (H2O+) to enable full utilization of real-world offline datasets and imperfect simulators for cross-domain policy learning. Our method addresses several key weaknesses in the previous method H2O, and offers flexibility to bridge various choices of strong offline RL backbones without introducing excessive conservatism. Through extensive simulation and real-world experiments, we show that our method outperforms the SOTA cross-domain RL methods in a wide range of dynamics gap settings. This makes H2O+ an ideal candidate for many real-world tasks without high-fidelity simulators and sufficient offline data.

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